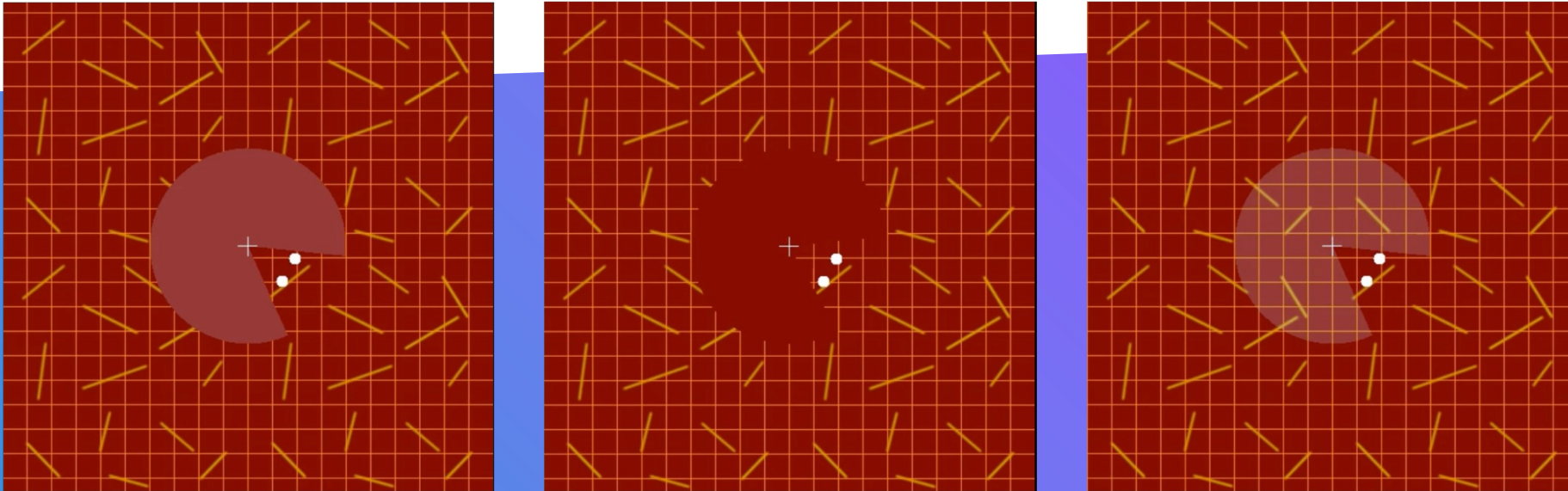


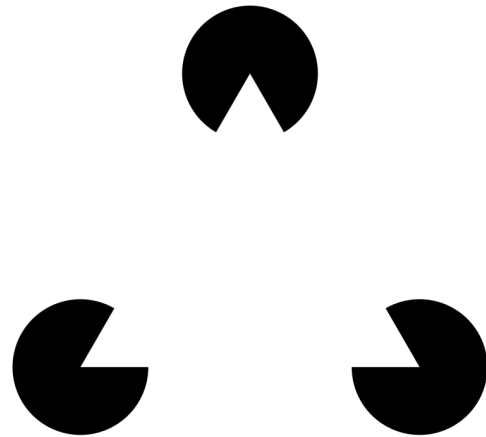


Investigating Partial Modal Completion Under Occlusion

The effect of occluder rotation speed on perceptual completion

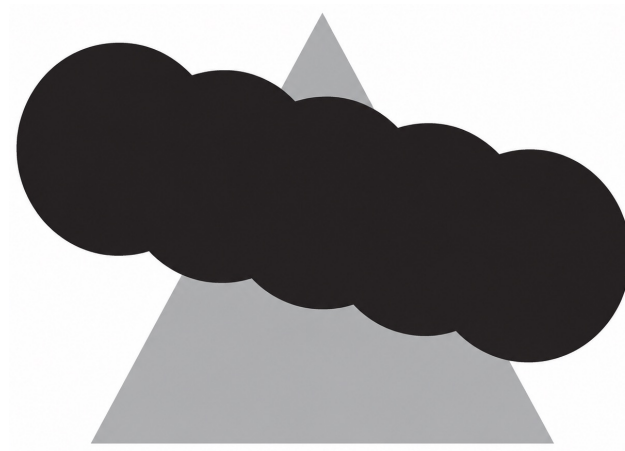
Mahdi Hassani · TU Berlin · Faculty IV — Supervisor: Dr. Guillermo Aguilar · July 2026

The perceptual illusion: seeing more than is physically given



Modal completion

Illusory contours or surfaces are experienced, although no physical contour is present.



Amodal completion

An object is perceived to continue behind an occluder, but the hidden part is not visibly filled in.

Traditional idea

Modal and amodal completion have often been treated as distinct categories: visible completion versus hidden continuation

Kanizsa (1976); Michotte et al. (1964/1991)

Partial modal completion

Palmer et al. (2007): the occlusion illusion



Two half-disks are physically **identical**. The one partially hidden behind an occluder can appear less occluded— so it looks **larger**.

A static occluded shape can appear partly “completed” into the visible region.

Palmer, Brooks, & Lai (2007), Perception.

Completion as a spectrum

Scherzer & Ekroll (2015)

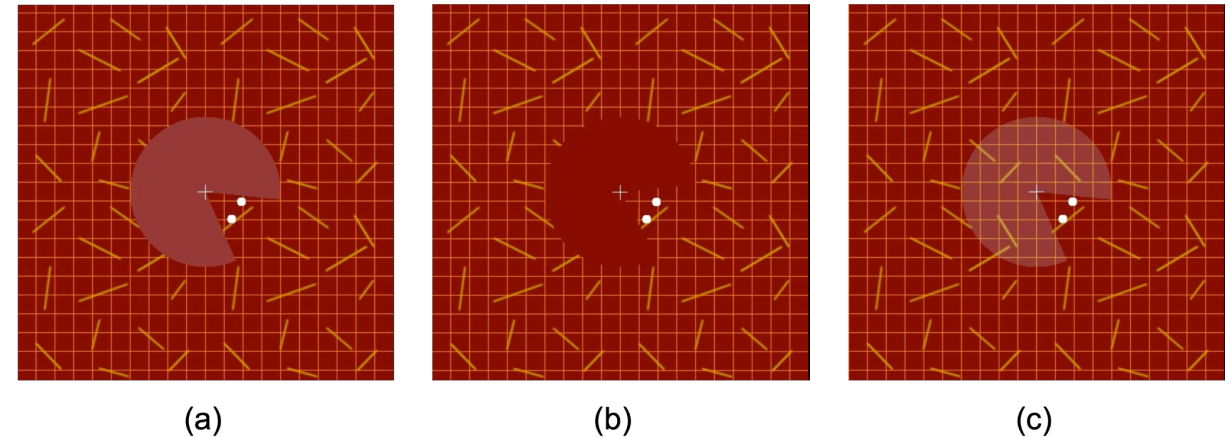
CONCLUSIVENESS OF EVIDENCE →

Amodal

Modal

How **vividly** we perceive a hidden part depends on the **conclusiveness of the sensory evidence hypothesis**.

Stronger evidence pushes perception from merely inferred toward genuinely seen.

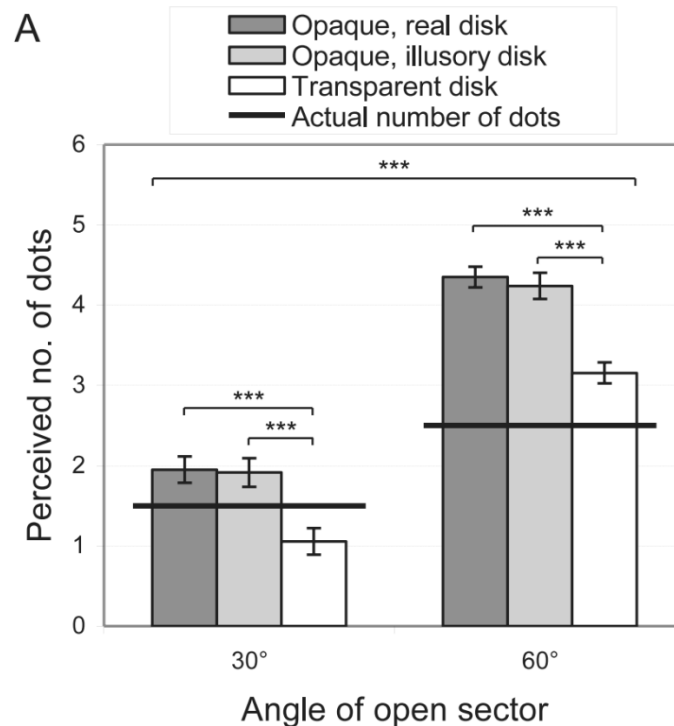


- A rotating occluder with an open sector moved in front of a ring of white dots.
- At any instant, only the dots inside the gap were physically visible.
- The experiment used different occluder types and gap sizes, especially 30° and 60°.

Scherzer & Ekroll (2015); Scherzer & Faul (2019)

Completion as a spectrum

Task: Observers were asked to report how many dots they saw through the gap.



Result

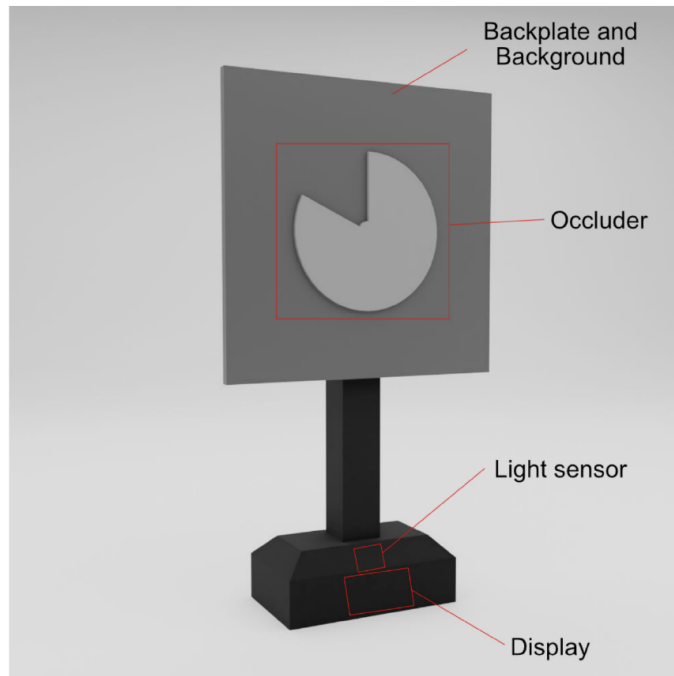
- With a **30° gap**, only about **1–2 dots** were physically visible, but observers reported about **2 dots**.
- With a **60° gap**, only about **2–3 dots** were physically visible, but observers reported up to **4.5 dots**.

Interpretation

- The rotating gap provided strong evidence that a complete ring of dots existed behind the occluder.
- This made some hidden dots contribute to visible perception, supporting the idea of partial modal completion under dynamic occlusion.

Scherzer & Ekroll (2015)

Haase (2025): bachelor thesis in a real-world setup



5

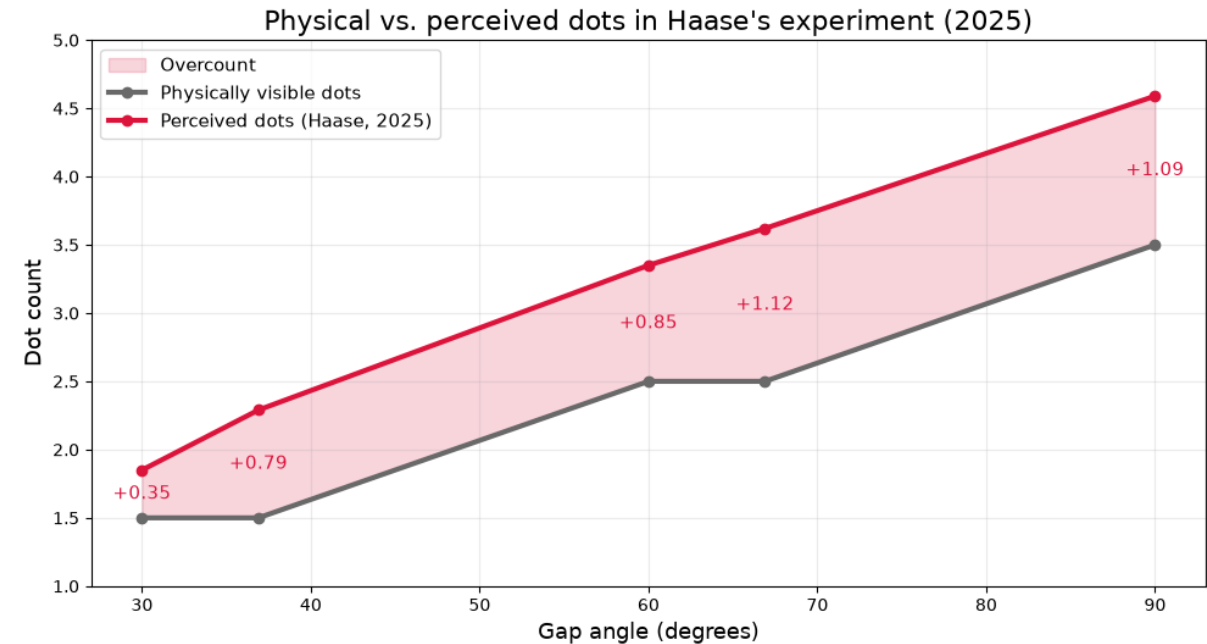
gap sizes

1

rotation/s

11

participants



- Real-world replication with a motorized apparatus.
- Perceived dot counts were above the physically visible dots; the shaded area shows the overcount.
- **Both studies used a fixed rotation speed; the effect of speed remains open.**

Data from Haase (2025)

Hypothesis & Research question



Working hypothesis

- If conclusiveness depends on temporal evidence, speed should matter.
- Faster rotation may provide more frequent and continuous evidence for hidden dots, increasing perceived completion.

slow

longer viewing of each dot

medium

reference speed from prior work

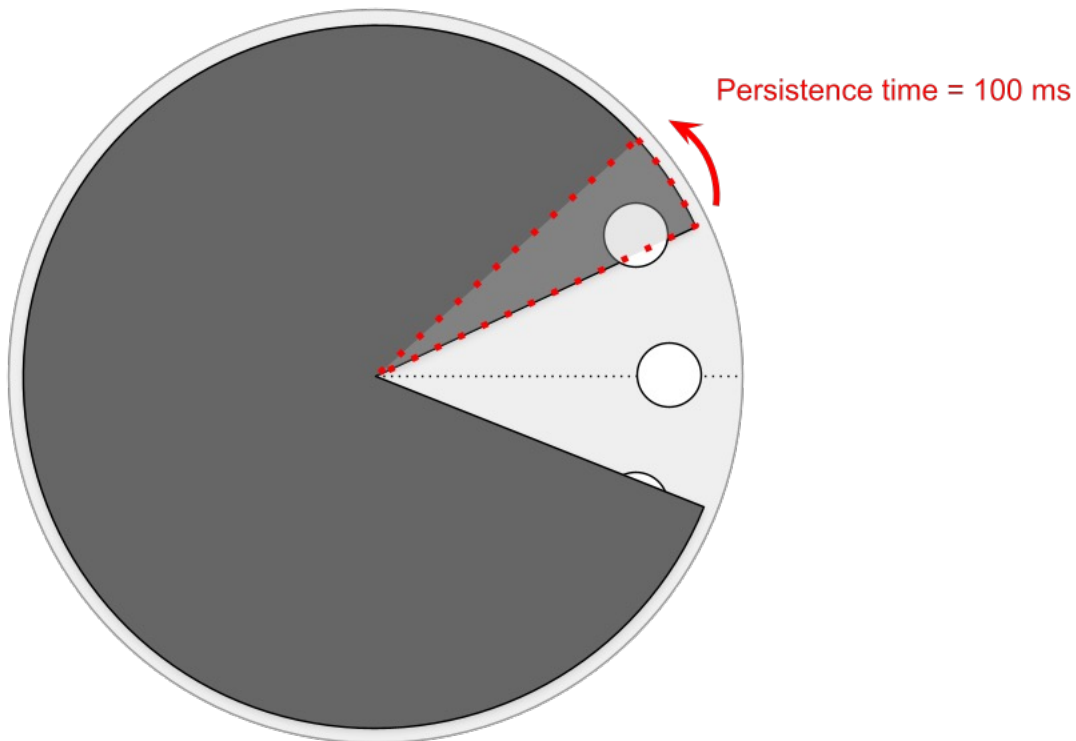
fast

more frequent reappearance

Research Question

How does the occluder's rotation speed influence the conclusiveness of sensory evidence and the perceived number of dots?

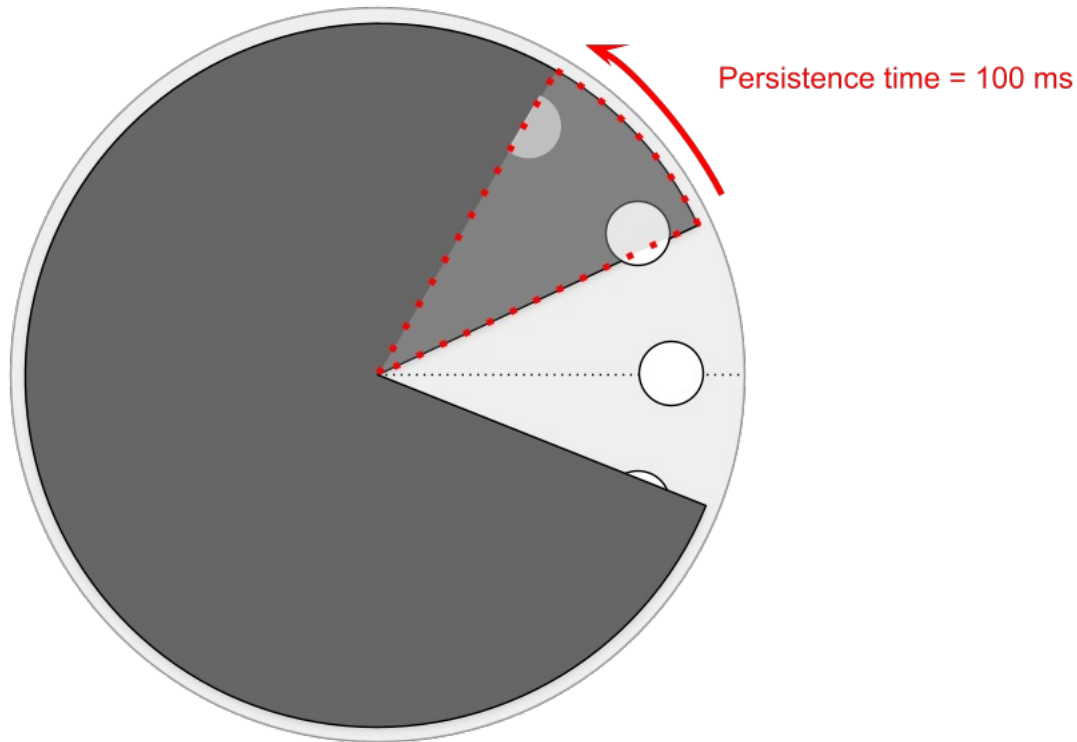
Model intuition: overcount as an effective gap extension



If a dot remains briefly represented after it is covered, it may still contribute to the percept. The visual system may therefore behave as if the opening were slightly larger than its physical size.

Persistence concept: Coltheart (1980)

Model intuition: overcount as an effective gap extension



- Persistence time is assumed to remain constant.
- Faster rotation means the occluder covers more angular distance during that same time.
- This increases the effective extension of the gap.
- As a result, more recently hidden dots may still contribute to the percept.

effective gap = physical gap + persistence extension

$$\Delta g = 360^\circ \cdot w \cdot \tau$$

w = rotation speed, τ = fixed persistence time

Persistence concept: Coltheart (1980)

Planned experiment: same apparatus, speed as the new factor

Experimental factors

5

gap sizes

3

speeds

5

repetitions

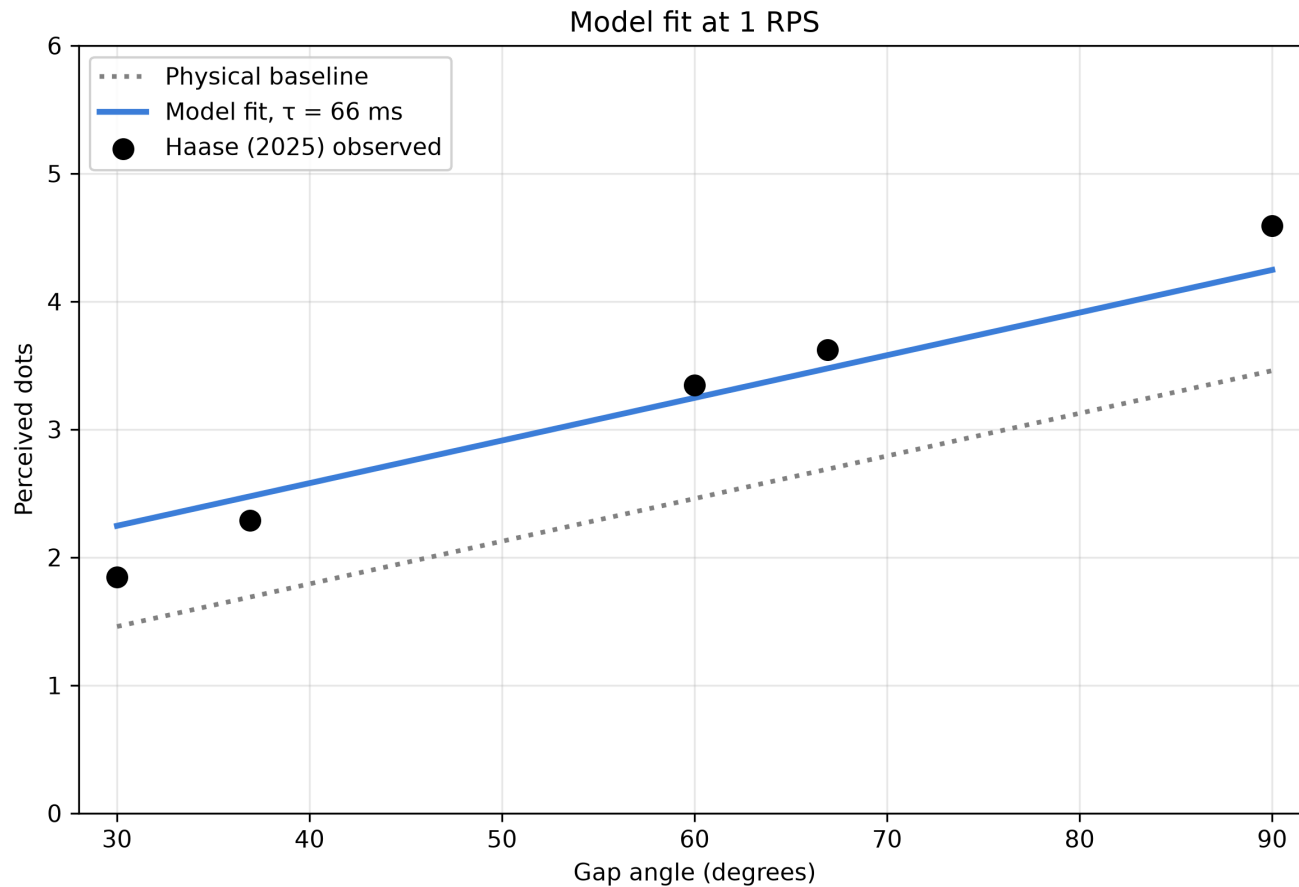
= 75 trials per participant

- Gap sizes: 30°, 36.9°, 60°, 66.9°, 90°
- Speeds: approximately slow, medium, fast (0.5, 1.0, 2.0 RPS)
- Dependent variable: perceived number of dots visible through the gap
- Trial order randomized for each participant

Goal

Isolate whether changing temporal evidence changes perceptual completion.

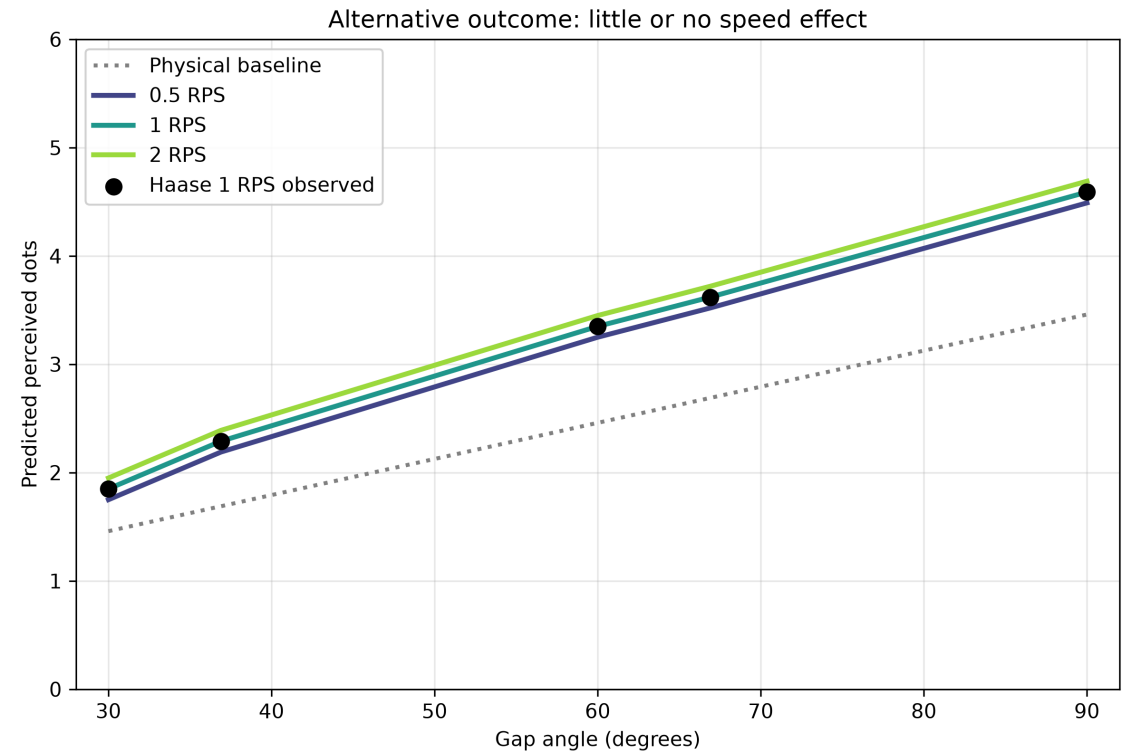
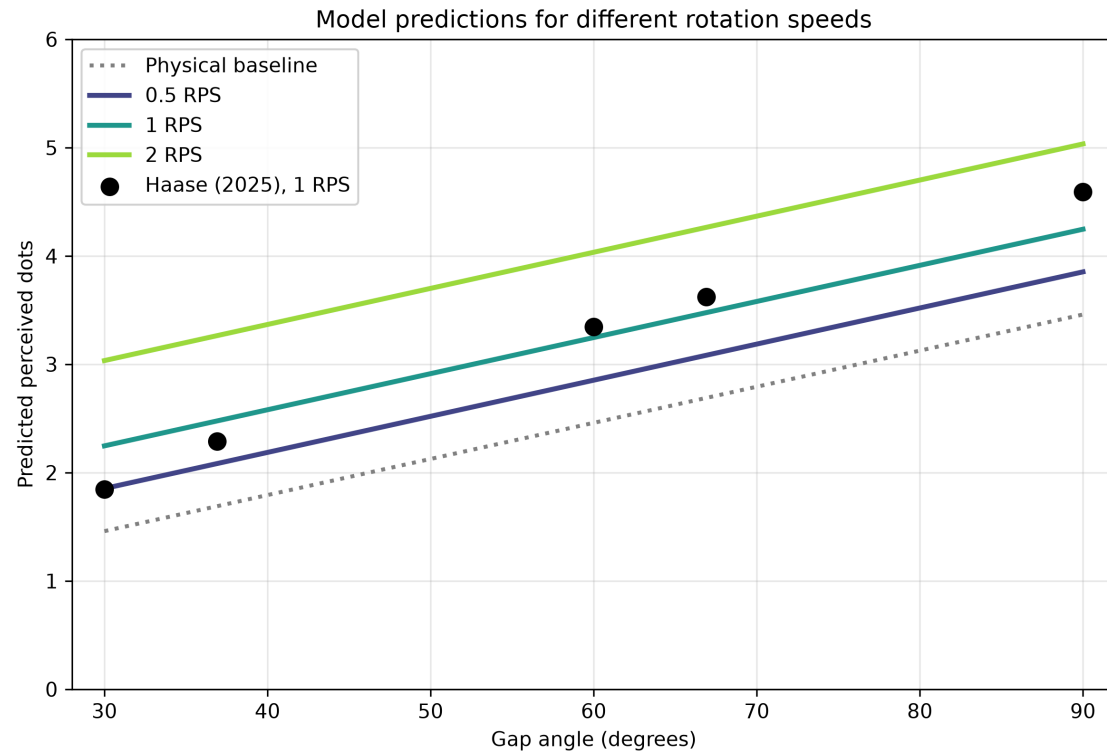
Linear fixed-persistence model: fitting Haase's data



- The model is fitted to Haase's 1 RPS data.
- With the data from Haase's experiment, the fitted persistence time is $\tau \approx 66$ ms.
- The fitted value provides a simple reference for speed predictions.

Haase (2025); Coltheart (1980).

Two possible outcomes: does speed matter?



Planned procedure and analysis



Statistical plan

- Average perceived dot count for each participant and condition.
- Compute illusion magnitude: perceived count minus physical baseline.
- Test effects of speed and gap size with repeated-measures analysis.
- Compare the empirical speed pattern with the fixed-persistence prediction and the no-speed-effect alternative.

Next steps

- Pilot test instructions and trial duration
- Collect participant data
- Run repeated-measures analysis
- Compare empirical data with model predictions
- Write results and discussion

Questions?



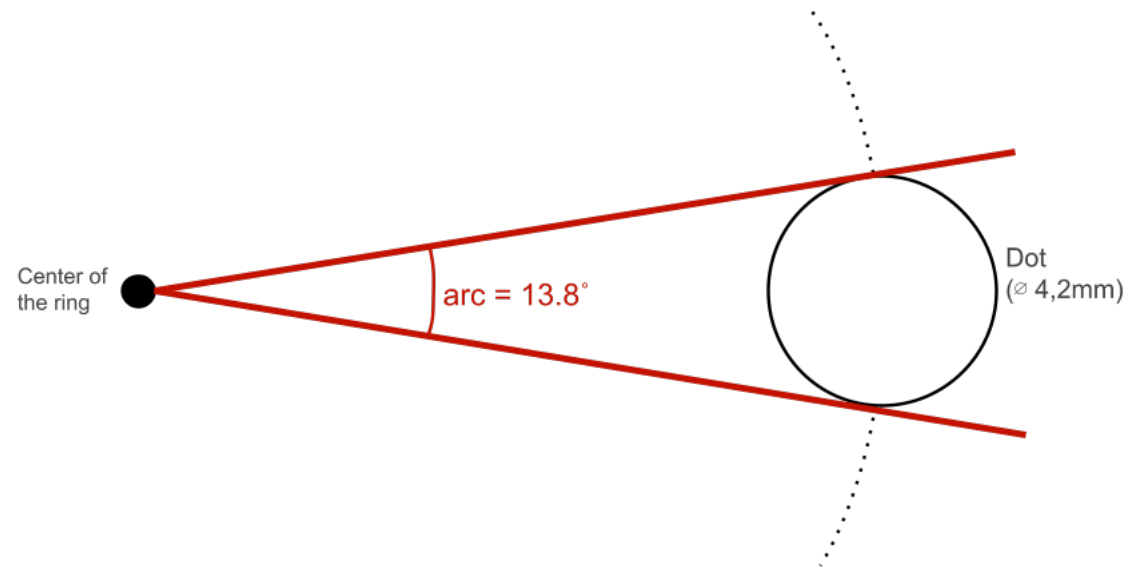
References:

- Kanizsa (1976)
- Michotte & Burke
- Palmer et al. (2007)
- Scherzer & Ekroll (2015)
- Scherzer & Faul (2019)
- Haase (2025)
- Coltheart (1980)

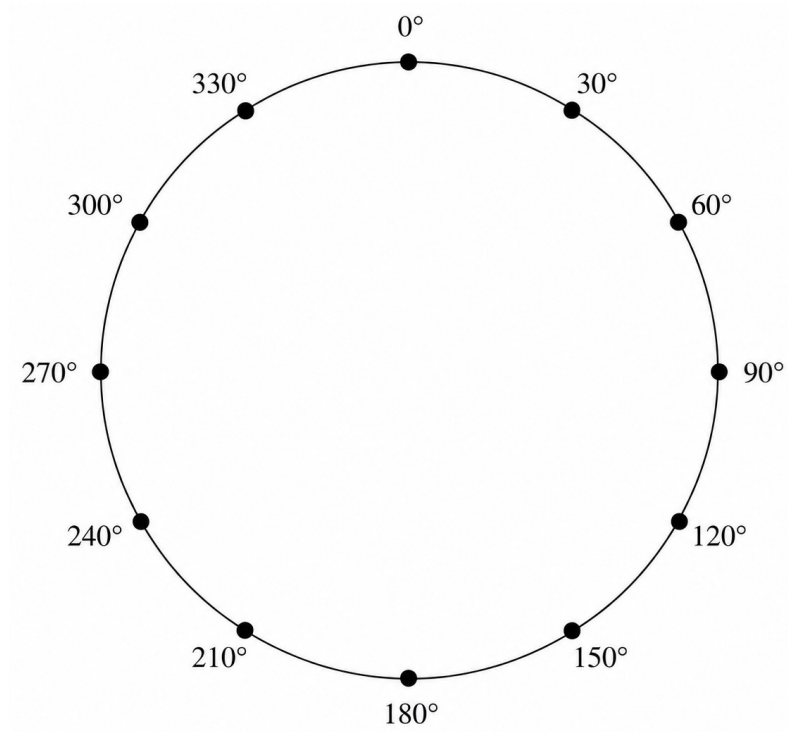
Backup slides



Arc calculation



Dot positions around the ring



Motor Issue: limited frequency with PWM API



Different requested speeds collapsed into a small set of supported PWM frequencies.

`gpioSetPWMfrequency`

Observed mapping

Requested speed	Requested STEP freq.	Actual API output
0.47 RPS	752 Hz	800 Hz
0.94 RPS	1504 Hz	1600 Hz
1.88 RPS	3008 Hz	4000 Hz

Result: the high-speed condition was not 1.88 RPS; it was effectively $4000 / (200 \times 8) = 2.5$ RPS.

Solution: use hardware PWM



`gpioHardwarePWM`

Before vs after

Speed	Old actual	New intended
0.47 RPS	800 Hz	752 Hz
0.94 RPS	1600 Hz	1504 Hz
1.88 RPS	4000 Hz	3008 Hz

Takeaway: hardware PWM removes the frequency quantization problem, so the speed condition matches the motor command.